



GE Healthcare

Technical Publications

Revision 7.1

DISCOVERY MR750™ and W450

(MR Main Disconnect Panel)

Installation and Service Manual

Reference 5161287 R9, M7000WL

For GE HEALTHCARE

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GE (USA) CONFIDENTIAL AND PROPRIETARY

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Troubleshooting and servicing should be performed only by a qualified electrician.



AC and DC control power is present when the breakers are in the open position. De-energize incoming power and lockout before servicing this panel.

This panel has been wired for use at 480V/60HZ; the panel can be converted for use at 380-400-415V/50-60HZ by changing the two transformer primary wiring connections from 480V to 400V. Refer to section 2-3.

1- Theory of operation

This panel incorporates a number of features desirable by MR installations to minimize down time, protect PDU (Power Distribution Unit) electronics, reduce operating costs, and reduce operational delays after a power outage. The panel comes wired with a single main circuit breaker and individual branch breakers for the PDU and Heat Exchange Cabinet (HEC).

The PDU branch circuit is controlled by a solid state circuit breaker with undervoltage trip, which opens the PDU circuit breaker on any loss of power. The PDU under-voltage trip circuit breaker CB2 protects the PDU's sensitive electronic equipment from sags and surges which immediately follow power loss from blackouts, storms, utility reclosure operations and out of phase automatic transfer switch operations by tripping the PDU circuit breaker on any loss of power. UV trip specifications are the UV trips the circuit breaker on a 30-70% loss of voltage. Voltage must be present on the circuit breaker two UV blue wires in order to reset the circuit breaker and turn ON the circuit breaker.

The Heat Exchange Cabinet (HEC) branch circuit utilizes an automatic restarting control circuit, which restores power after a power outage. A 12V DC relay CR1 controls the operation of the HEC branch circuit contactor. Operation of the 12V DC power supply and battery provide the control power for DC relays CR1 and CR2. The panel has a HEC contactor, which is controlled by the DC auto start control circuit. Relays CR1 and CR2 must both be energized in order for the PDU circuit breaker under-voltage trip and the HEC branch circuit contactor, C1 to close. DC battery control circuit protection time is based upon the condition of the battery but is expected to hold for at least 48 hours. The DC Battery has a life of 2 years at which time the battery and charger must be replaced to maintain auto-restart capability. Semi annual verification on the charger and battery are required.

PDU and HEC power Emergency Off disconnection is also provided by this panel. Two normally closed remote emergency off pushbuttons are included with this panel for wall mounting. The cover-mounted power off pushbutton or either of the remote emergency off pushbuttons provide immediate shutdown of the entire system. If required for a particular installation, additional remote emergency off pushbuttons may be ordered from BEVCO Engineering, 262-820-2400. Remote emergency off pushbutton catalog numbers are listed at the end of this manual.

Restoration of complete power after an emergency off operation is accomplished by first pressing the System ON pushbutton on the cover of the panel and subsequently resetting the PDU circuit breaker by moving the PDU circuit breaker handle to OFF then to ON. Circuit breaker CB3 provides power for the HEC as indicated on the attached wiring diagram.

The power supply green indicating indicates the presence of DC voltage either from the power supply or from the battery.

1-1 SPECIFICATIONS

1-1-1 ELECTRICAL SPECIFICATIONS

UL design and ratings require the application of wire at the 75-degree ampacity ratings of NEC table 310.16. Wire rated at higher temperature ratings such as 90 degree may be used but ONLY applied at the 75-degree ampacity ratings of NEC Table 310.16. All circuits require a green or green/yellow ground conductor.

UL, cUL labeled and CE labeled. RoHS Compliant

Power system: 3 Phase, Ground. If neutral is utilized it must be terminated in this panel.

Main Feed wire sizes;

Main circuit breaker #8 - 350KCMIL maximum.

Main Ground Bus #6 - 250KCMIL maximum.

Main Neutral Bus #4 – 350KCMIL maximum.

480V/60HZ, 380-400V/50HZ, 3 phase 3 wire + ground.

Panel Short Circuit Current Rating is 25,000A @ 480V

Main fuses are provided for Short Circuit Current protection. These fuses are oversized at 200 amperes for the PDU circuit and 60 ampere for the HEC cabinet to provide back-up protection.

Circuit Breaker CB2 utilizes 200A rated fuses for the PDU

Circuit Breaker CB3 utilizes 60A rated fuses for the Heat Exchanger cabinet

380, 400, & 415Volt, 50/60HZ

Circuit Breakers CB1, CB2 and CB3, UL 489 & CSA C 22.2

Circuit Breaker Interrupting Rating 25,000 AIC Symmetrical 50/60 Hertz

Panel Short Circuit Current Rating is 25,000 Amperes RMS symmetrical at 480 volt AC.

IEC 947-2 Interruption Capacity

SFHA Main Circuit Breaker 380-415V, Ue, Icu 35,000 RMS Symmetrical AMPS 50/60 Hertz.

SFHA Main Circuit Breaker 380-415V, Ue, Ics 17,000 RMS Symmetrical AMPS 50/60 Hertz.

SEHA Branch Circuit breakers 380-415V, Ue, Icu 15,000 RMS Symmetrical AMPS, 50/60 Hertz

SEHA Branch Circuit breakers 380-415V, Ue, Ics 10,000 RMS Symmetrical AMPS, 50/60 Hertz

1-1-2 ENVIRONMENTAL SPECIFICATIONS

FOR INDOOR USE ONLY

Temperature 59-95F (15-35C)

Humidity 30-75%, NON-CONDENSING

COVER LABELS

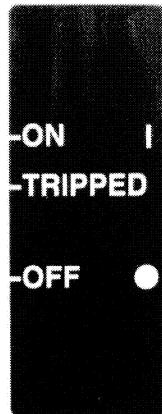


Figure 1-1-3-1
Circuit Breaker plate

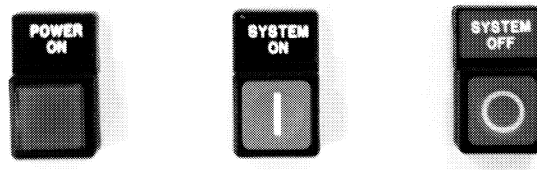


Figure 1-1-3-2 Pushbutton plates

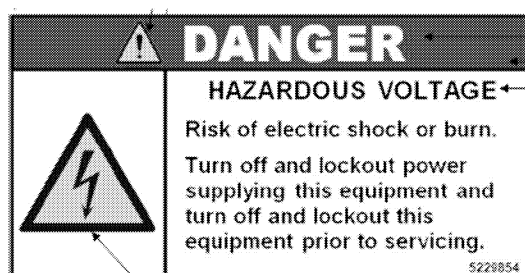


Figure 1-1-3-3
Danger shock label

**This Panel is fed from the
source indicated below.**

Panel _____ Ckt _____

Figure 1-1-3-4
Source label

INTERIOR LABELS

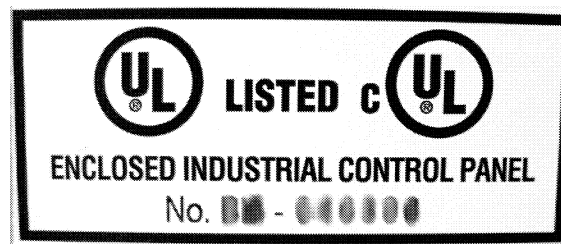


Figure 1-1-3-5
Nationally recognized test Laboratory approval label

Battery Maintenance label

BATTERY MAINTANENCE RECORD

ANNUAL INSPECTION AND TESTING OF THE MDP
BATTERY IS REQUIRED. THE BATTERY SHALL BE
REPLACED AFTER TWO YEARS OF SERVICE.
ORIGINAL INSTALLATION

DATE _____ BY _____

ADD DATE, CROSS OUT "CHECKED OR REPLACED" TO
INDICATE SERVICE PERFORMED AND SIGN

DATE _____ CHECKED/REPLACED BY _____

DATE _____ CHECKED/REPLACED BY _____

DATE _____ CHECKED/REPLACED BY _____

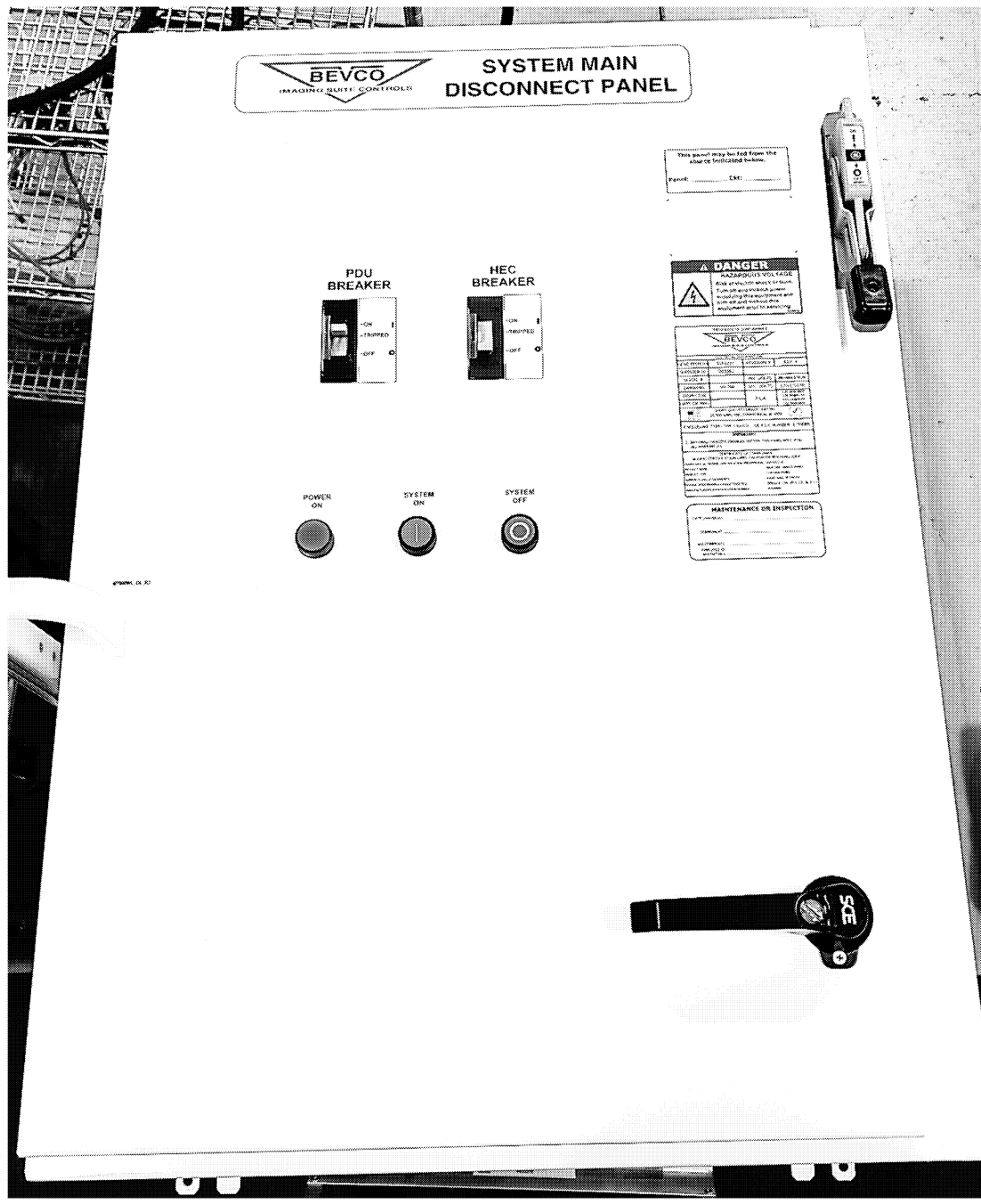
DATE _____ CHECKED/REPLACED BY _____

DATE _____ CHECKED/REPLACED BY _____

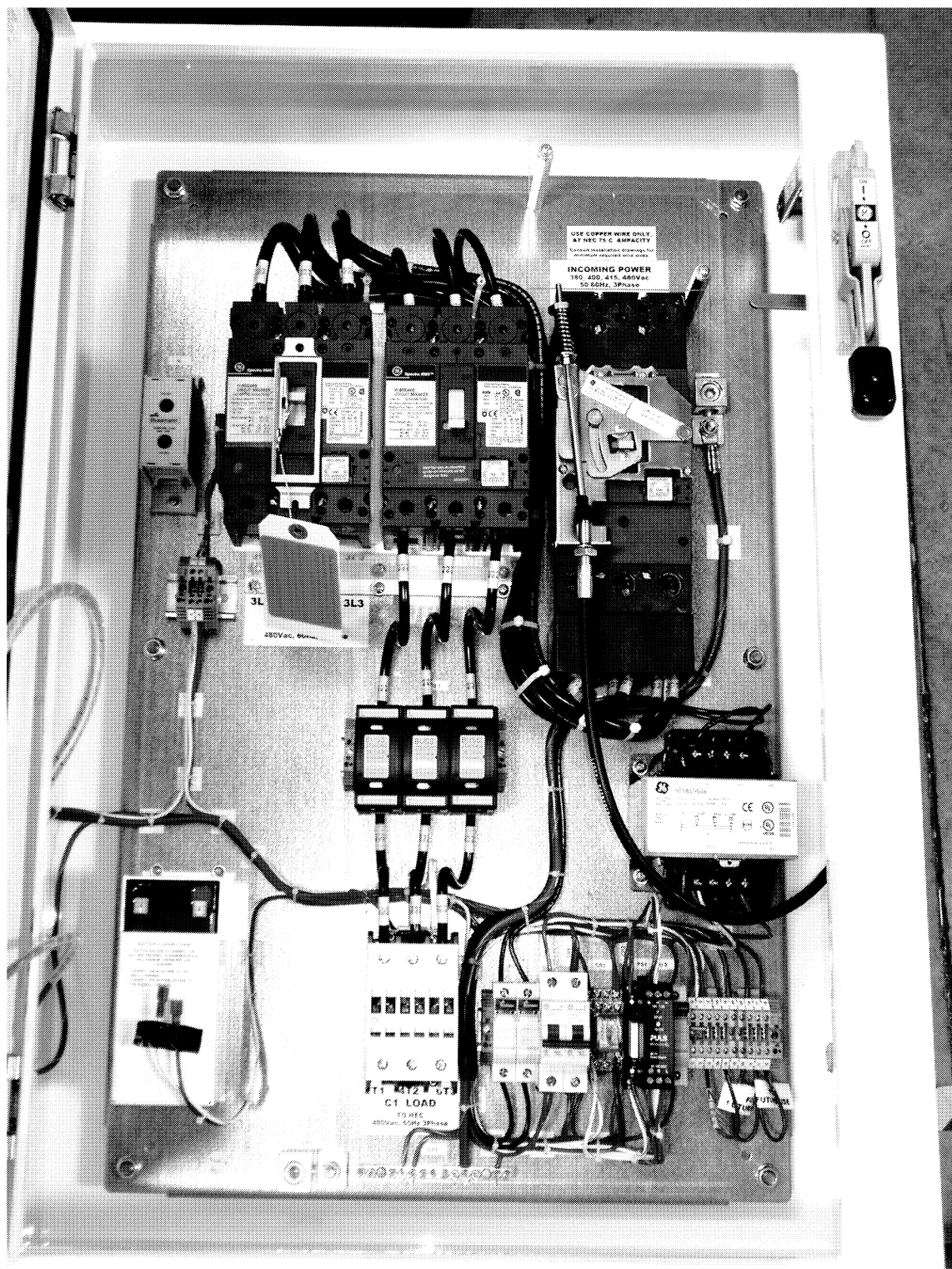
DATE _____ CHECKED/REPLACED BY _____

DATE _____ CHECKED/REPLACED BY _____

FRONT COVER PHOTOGRAPH 1-1-4-1



INTERIOR PHOTOGRAPH 1-1-4-2



1-2 Panel Dimensions

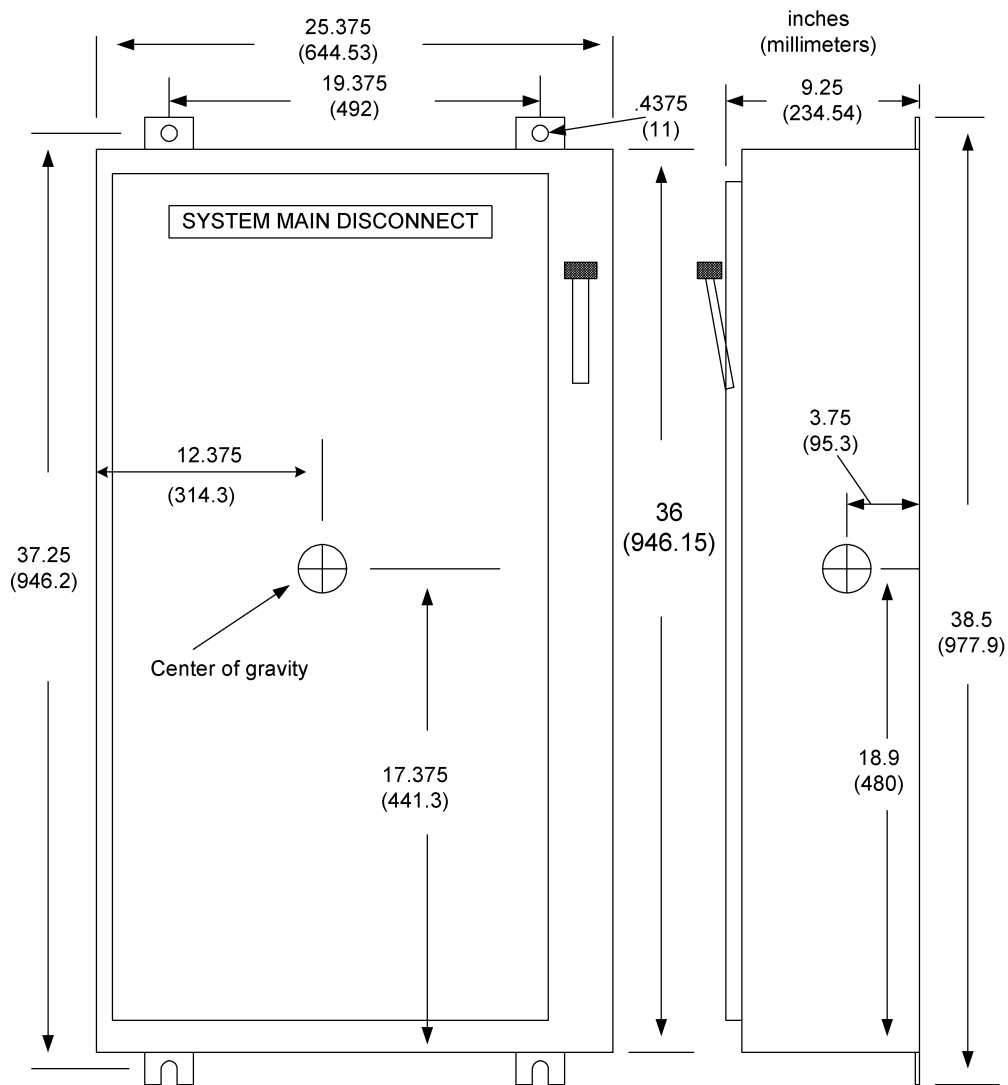
The panel may be recessed approximately 6.95 inches (176.5 mm) into the wall for semi-flush installations.

TABLE 1-1
PANEL EXTERNAL PHYSICAL DIMENSIONS (approximate)
Verify dimensions before installing

PARAMETER	DIMENSIONS	
Height-Enclosure	36.0 inches	914.4 mm
To edge of mounting flanges	38.5 inches	977.9 mm
Width	25.375 inches	644.525 mm
Depth	9.25 inches	234.54 mm
Weight	150 lbs	68.4 kg
Door Swing Radius	23.5 inches	596.9 mm
Center of Gravity as indicated below.		

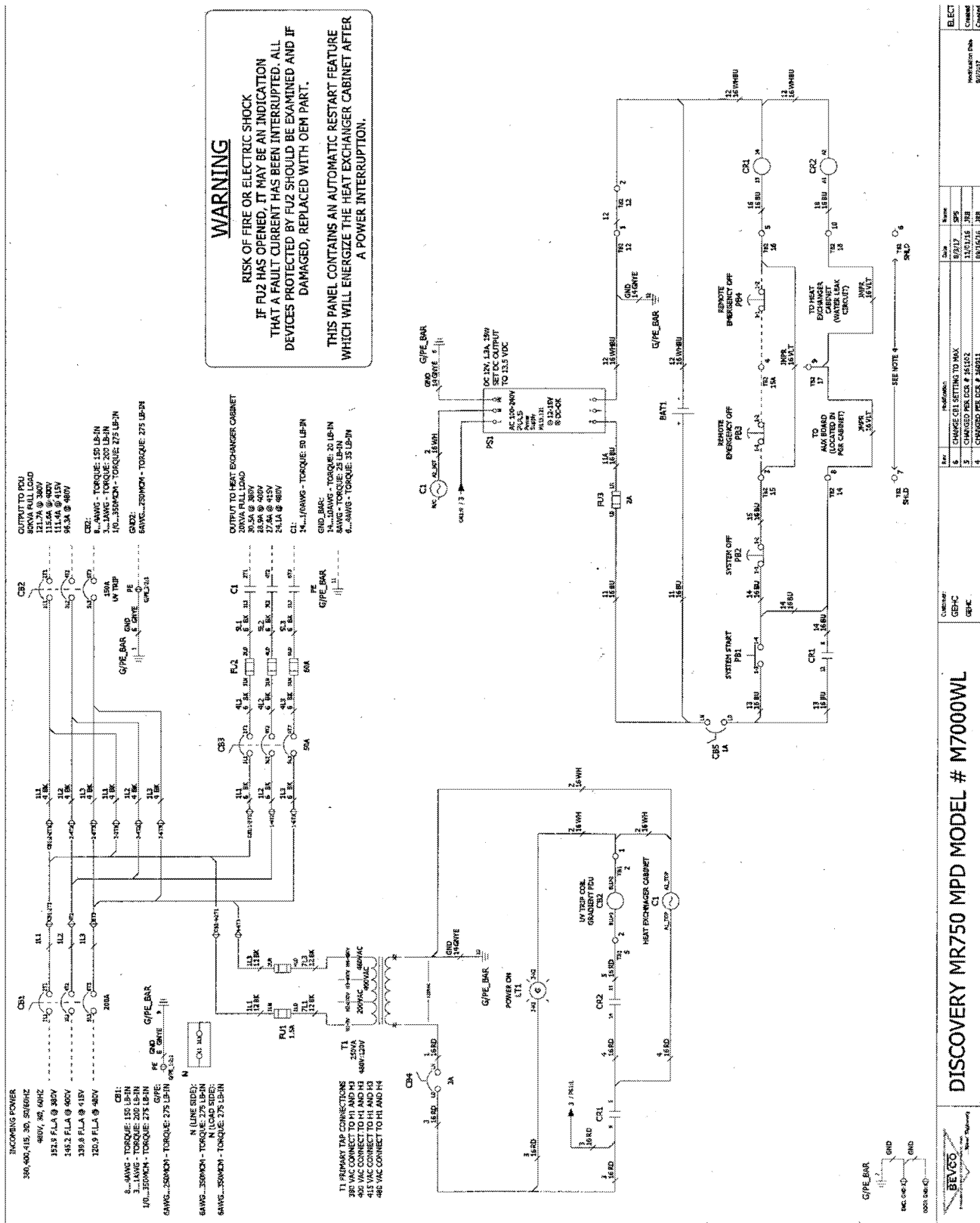
FRONT VIEW

MOUNTING PATTERN



FRONT AND BACK VIEW OF MAIN DISCONNECT PANEL
ILLUSTRATION 1-1

1-3 Electrical Wiring Diagram



MAIN DISCONNECT PANEL WIRING DIAGRAM ILLUSTRATION 1-3

CIRCUIT BREAKER INSTANTANEOUS ADJUSTMENTS

Fuse ratings and CR1 user remote indication contacts

CB1 INSTANTANEOUS TRIP SETTING					
MIN	2	3	4	5	6
3X	3.8X	4.8X	6X	7.8X	10X
590	750	940	1180	1535	2000

SET

CB2 INSTANTANEOUS TRIP SETTING					
MIN	2	3	4	5	6
3X	3.8X	4.8X	6X	7.8X	10X
4440	560	705	885	1150	1500

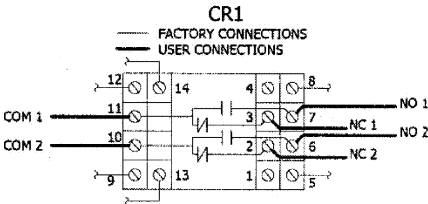
SET

CB3 INSTANTANEOUS TRIP SETTING					
MIN	2	3	4	5	6
3X	3.8X	4.8X	6X	7.8X	10X
148	187	236	296	386	498

SET

FUSE REPLACEMENT CHART					
FOR CONTINUED PROTECTION AGAINST RISK OF FIRE, REPLACE FUSES WITH SAME TYPE AND RATING BELOW.					
FUSE	SIZE	CLASS	LITTLE FUSE	MERSEN	BUSSEMAN
FU1	1.5A	CC	KLD01.5	ATQ61-1/2	PHQ-6-1-1/2
FU2	60A	J	JTD600JD	ATT60	LPJ-60SP
FU3	2A	CC	CCMR002	ATOR2	LP-CC-2

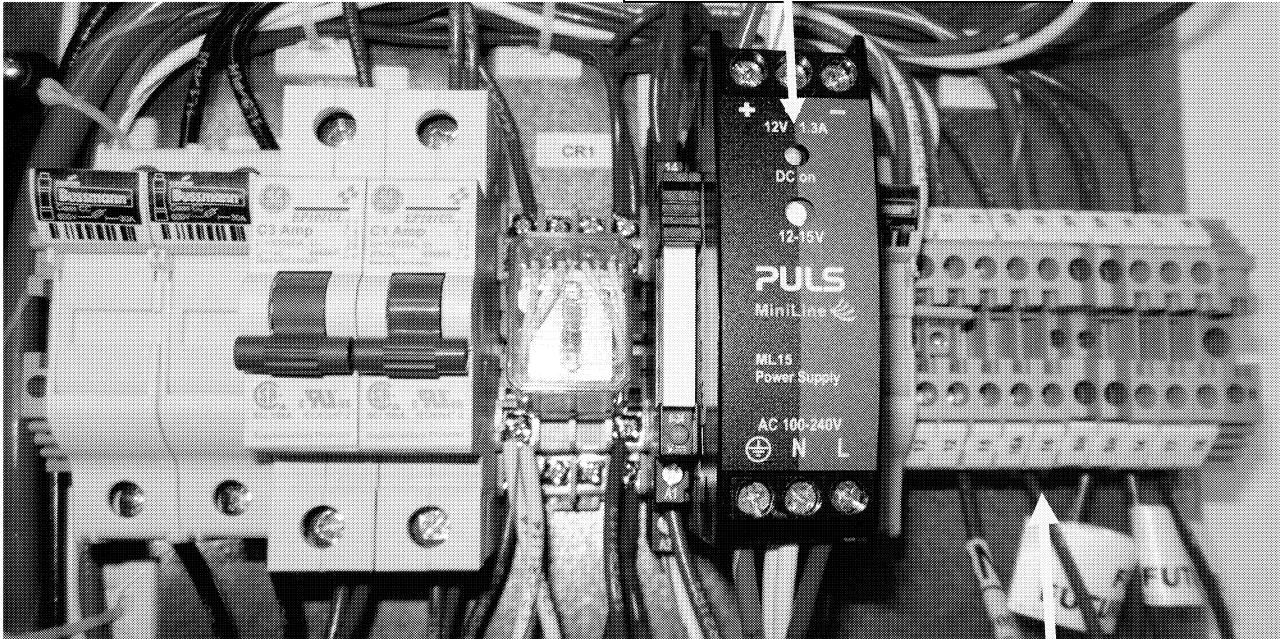
TO KEEP ROHS COMPLIANT, FU3 MUST BE CCMR002.



SHORT CIRCUIT CURRENT RATING:
25,000A RMS SYMMETRICAL @ 480V
ROHS COMPLIANT

1-4 Control terminals

Relay CR2 green light must be illuminated for the PDU circuit breaker to be reset and closed.



CONTROL TERMINAL STRIP ILLUSTRATION 1-4

Customer control connections are located on this side of terminal blocks

REMOTE EPO CONNECTIONS

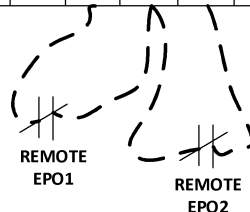
EXISTING TB2 CONNECTIONS

13	14	15	15A	16	17	18
----	----	----	-----	----	----	----

READ LABEL ON THE VIOLET WIRE.
REMOVE VIOLET WIRE BETWEEN
TB2-15 AND TB2-16 when installing
remote EPO pushbuttons.

REMOTE TB2 EPO CONNECTIONS

13	14	15	15A	16	17	18
----	----	----	-----	----	----	----



CONNECT NORMALLY CLOSED EPO PUSHBUTTONS AS INDICATED ABOVE

2 INSTALLATION

2-1 Location

- Review the MR Suite Magnetic Field Specifications usually section 7 of the MR Product Installation Manual before installing the Main Disconnect Panel (MDP).
- The Main Disconnect panel should not be located within the area of significant magnetic field as the magnetic field may affect the operation of the internal electrical components.
- The Main Disconnect panel contains a transformer and circuit breakers which may have an effect on the MR images.
- The installer should review the mounting provisions of the MDP and securely mount the panel in accordance with good engineering practice and all local requirements.

2-2 Installation

- Incoming 3 phase 3 wire power is connected to the main breaker located at the top of the Main Disconnect Panel. Maximum conductor size is #8-350KCMIL.
- Incoming ground is terminated to the main ground lug.
- Confirm if the MR system requires a neutral as many do not require a neutral conductor. If neutral is existing it may be terminated in the main disconnect neutral block.
- If larger conductors are utilized then the larger conductors are terminated in a junction box above the MDP and a 4/0 - 350KCMIL conductors are spliced in the junction box and terminated to the Main CB, CB1. The 200A rating of the upstream overcurrent device will protect the 4/0-350KCMIL conductors, which are rated 310 amperes as well as the larger conductors.
- **All circuits require a green or green/yellow ground conductor.**
- There are no knockouts in the panel for conduits so the installer can punch the knockouts where they prefer to have them.
- All conductors must be sized per the GEHC wiring tables on the installation drawings or the GEHC system installation manual. The PDU is connected to the load side of fuse block FU2, terminals 3L1, 3L2, and 3L3. Conductor range is #6 AWG to 250KCMIL conductors.
- Heat Exchange Cabinet (HEC) connections are made at contactor C1 terminals 2T1, 4T2, and 6T3. Conductor size is #6AWG- 1/0 AWG.
- Verify that the DC power supply output voltage with the battery blue wire disconnected is 13.55V. The potentiometer on the power supply can be used to adjust the PS output voltage if necessary.
- Carefully, connect the battery wires to the 12VDC battery. Blue wire number 11 is connected to the positive terminal with the red paint. Wire number 12, (white with blue stripe) is connected to the negative terminal with the black paint.
- A VIOLET temporary jumper wire is installed in the factory between TB2-15 and TB2-16 in place of the remote EPO pushbuttons. Removal of the violet jumper wire when connecting the remote EPO pushbuttons.
- One remote EPO pushbutton is wired to TB2-15 and TB2-15A and the second remote EPO is wired to TB2-15A and TB2-16.
- The green power supply light indicates the presence of DC voltage either from the power supply or the battery. 10.5 volts is the voltage required to light the green light.
- PDU circuit breaker has an undervoltage trip coil, which must be energized to enable the PDU circuit breaker to be reset or closed. Any loss of 120V undervoltage power will trip the PDU circuit breaker.
- Relays CR1 and CR2 control 120V coil power to CB2 undervoltage trip. Relay CR2 has a green indicating light illuminated when it has been energized with its 12VDC control voltage. To trouble shoot CR1 or CR2 verify that CB5 is closed, system on pushbutton has been pressed energizing CR1 and there is 12VDC to ground from terminals 14, 15, 16, 17 and 18.

- The circuit breaker undervoltage trip coil is fed from the 2 blue wires exiting CB2 from the left side of the circuit breaker.
- Mounting height of the centerline of the main breaker or the handle of the operating mechanism when in the ON position must not exceed 6ft-7in. (2m) above the floor per National Electric Code (NEC # 404.8).
- Complete the label on the cover of the panel indicating the location and circuit of the power source providing power to this panel.

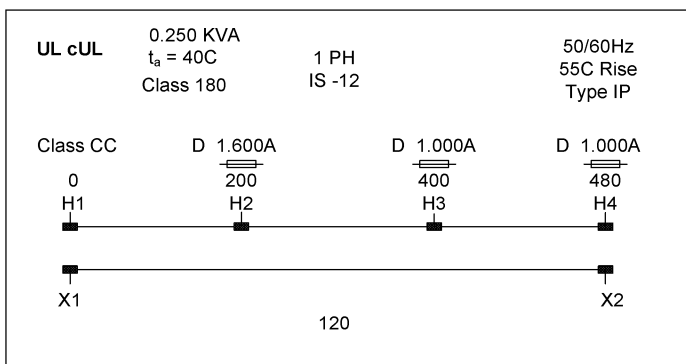
The control circuit wiring must be completed by installing the two remote emergency off pushbuttons as shown on the wiring diagrams on the inside of the door. The PDU circuit breaker and HEC contactor will not operate properly unless the remote emergency off pushbuttons are installed and the factory installed temporary jumper removed.

- ♦ **For parts or assistance contact BEVCO Engineering, 262-820-2400, Central time zone, Fax 262-820-2599.**

2- 3 Conversion to 400V for 380V, 400V or 415V, 50Hz operation.

This panel is suitable for operation at the above stated voltages with the changing of the control transformer primary tap setting.

250VA Control Transformer CPT1 480V to 400V Conversion



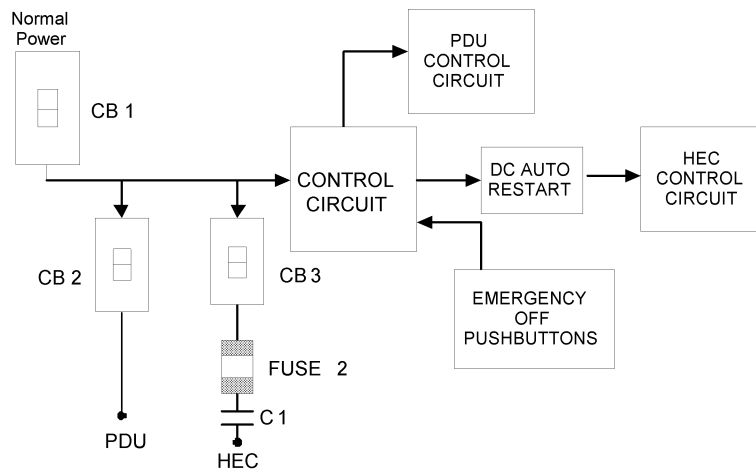
1. Open main circuit breaker CB1.
2. Open two-pole fuse holder FU1.
3. Remove wire originally connected to transformer terminal H4.
4. Connect wire from Step 2 to transformer terminal H3.
5. Close fuse holder FU1.

NOTE: Transformer 200V tap, H2, is not used in this application.

2-4 SYSTEM CONFIGURATION

All circuits require a green or green/yellow ground conductor.

See illustration 2-1 below for the System Configuration.



2-5 INCOMING POWER CONFIGURATION

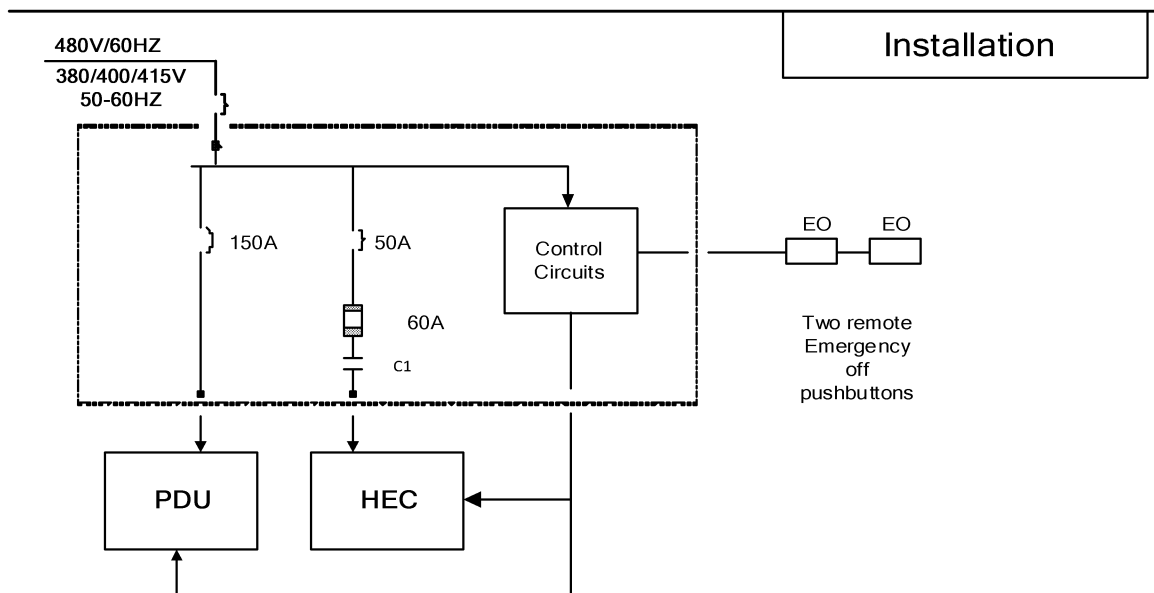
See illustration 2-2 for the incoming power configuration.

All circuits require a green or green/yellow ground conductor.

INCOMING POWER CONFIGURATION

ILLUSTRATION 2-2

All circuits require a green or green/yellow ground conductor.



- Fuses are oversized for backup short circuit protection

2-6 Panel Testing

TESTING ACTION

1. Close Main Circuit Breaker, CB1. Close Heat Exchanger circuit breaker CB3.
2. Press the System ON pushbutton and then reset PDU circuit breaker CB2. Verify the (HEC) Heat Exchanger Cabinet contactor C1 is ON.
3. Press any emergency off pushbutton.
4. Press the System ON pushbutton and then reset PDU circuit breaker CB2. Verify the (HEC) Heat Exchanger Cabinet contactor C1 is ON.
5. Press any emergency off pushbutton.

VERIFY

1. Green Power On light should light indicate system control power is available.
2. Verify with voltmeter that three-phase power is available for the PDU at Fuse FU2 terminals 3L1, 3L2, 3L3 and that Heat Exchanger Cabinet contactor C1 is closed. Verify three-phase power at contactor C1 terminals 2T1, 4T2, and 6T3. Green pilot light should be on indicating control power is available.
3. PDU circuit breaker CB2 must trip (open) and HEC contactor C1 must de-energize and stay de-energized. Verify with a voltmeter that the HEC power at the contactor C1 terminals 2T1, 4T2, and 6T3 is disconnected and the PDU power at Fuse FU2 lugs 3L1, 3L2, and 3L3 is disconnected at the fuse load terminals.
4. Verify with voltmeter that three-phase power is available for the PDU at Fuse FU2 terminals 3L1, 3L2, 3L3 and that Heat Exchanger Cabinet contactor C1 is closed. Verify three-phase power at contactor C1 terminals 2T1, 4T2, and 6T3. Green pilot light should be on indicating control power is available.
5. PDU circuit breaker CB2 must trip open and HEC cabinet contactor C1 opens.

TESTING ACTION

6. Starting with the de-energized state of #5, turn OFF incoming power at the panel main breaker CB1, feeding power to this panel.
7. With incoming power OFF from step 6, now restore main power to panel by turning panel main circuit breaker CB1 ON (switch to ON).
8. Press the System ON pushbutton and then reset PDU circuit breaker CB2. Verify the (HEC) Heat Exchanger Cabinet contactor C1 is ON and the green power on light is ON.
9. Press the System ON pushbutton. Verify the PDU circuit breaker is closed. Turn OFF main circuit breaker CB1 power and then turn on the main circuit breaker CB1 to restore power to the panel.

VERIFY

6. The green light must be OFF indicating loss of control power.
7. Upon restoration of power the HEC contactor C1 must remain OFF. This demonstrates that the emergency off function remains disabled during a loss of power. There should be no auto restart after any emergency OFF operation. The PDU circuit breaker was tripped in Step 5 and remains tripped.
8. The HEC contactor can be heard closing when the system ON push-button is pushed.
9. The green pilot light must be ON upon restoration of power. The PDU circuit breaker and HEC contactor should turn OFF with loss of power caused by opening main circuit breaker CB1. Only the HEC contactor C1 must turn ON with the restoration of power indicating the proper operation of the automatic restart feature.
The PDU circuit breaker CB2 turns off on any loss of power and remains off with restoration of power. PDU CB2 opens and does not close upon power restoration. The PDU does not automatically restart after a power loss. This demonstrates the DC supervisory circuit is operating properly and the HEC auto restart feature is operational.

WARNING!

AC AND DC CONTROL POWER IS PRESENT WHEN THE BREAKERS ARE IN THE OPEN POSITION. MAKE SURE NO ONE IS WORKING ON THE EQUIPMENT, WHICH THIS PANEL FEEDS, PRIOR TO TESTING THE PANEL.

3-TROUBLESHOOTING

Auto Restart function, PDU, and the HEC contactor will not operate properly unless the remote emergency off pushbuttons are wired as indicated on the wiring diagram and the factory installed temporary jumpers are removed.

Make sure the remote emergency off pushbuttons is wired through normally closed contacts.

PDU circuit breaker has an undervoltage trip coil, which must be energized to enable the PDU circuit breaker to be reset or closed. Loss of 120V undervoltage power will trip the PDU circuit breaker.

Relays CR1 and CR2 control 120V coil power to CB2 undervoltage trip. Relay CR2 has a green indicating light illuminated when it has been energized with its 12VDC control voltage.

If CR1 or CR2 does not operate verify that DC circuit breaker CB5 is closed, there is at least 12VDC to ground on the positive (+) terminal of the battery. Verify system ON pushbutton has been pressed and that there is at least 12VDC to ground from terminals 14, 15, 16, 17 and 18.

To trouble shoot CR2 verify that CB5 is closed, system on pushbutton has been pressed energizing CR1 and there is 12VDC to ground from terminals 14, 17, 18.

The circuit breaker undervoltage trip coil is fed from the 2 blue wires exiting CB2 from the left side of the circuit breaker.

Operation of any emergency off pushbutton should de-energize relay CR1, shutting the entire system down.

Check for broken or burnt out LED pilot lights.

Check to see that power supply wire labeled 11 is connected to the battery positive terminal marked with red paint. Check to see that power supply wire labeled 12 is connected to the battery negative terminal marked with black paint. 12V DC output voltage must be 12 to 13.55V DC under normal operating conditions. If DC voltage is less than 11V, replace the power supply and battery. 12V DC relay CR1 controls the operation of all branch circuit contactors. Operation of the 12V DC power supply and battery provide the control power for relay CR1. Relay CR1 must be energized in order for any branch circuit contactor to close.

The battery charger green indicates the presence of DC voltage either from the power supply or from the battery.

A VIOLET temporary jumper wire is installed in the factory between TB2-15 and TB2-16 in place of the remote EPO pushbuttons. Remove the violet jumper wire when connecting the remote EPO pushbuttons.

One remote EPO pushbutton is wired to TB2-15 and TB2-15A and the second remote EPO is wired to TB2-15A and TB2-16.

Main circuit breaker operator adjustments are normally not necessary. If the main circuit breaker operating mechanism requires adjustment it is paramount that you review the GE, GEH-6290 installation instructions before attempting adjustments.

The CB operator instructions are available from this link:

Refer to Cable Installation section, page 5, figure 7, paragraphs 5 to 7.

<http://apps.geindustrial.com/publibrary/checkout/GEH-6290?TNR=Installation%20and%20Instruction%7CGEH-6290%7CPDF&filename=GEH6290.pdf>

- ♦ ***For parts or assistance contact BEVCO Engineering, 262-820-2400, Central time zone, Fax 262-820-2599.***

4- MAINTENANCE

4-1 General Maintenance

Note

The auto restart circuit requires the 7 Ah DC battery to be functional. Semi-annually verify that the 12V DC charging circuit is operational a record of battery testing is located on the panel exterior door.

Annually, the unit should be serviced by a qualified electrician to tighten all electrical connections.

4-2 Battery and Power supply test

1. Open main circuit breaker and press the system on pushbutton to energize CR1 and CR2.
2. Measure the DC voltage at the battery quick connects terminals with the battery loaded. IF voltage is less than 11 volts or rapidly decreases battery replacement and charger replacement is necessary. Batteries are reliable for 2 years and should be replaced after 2 years.
3. Turn ON the main circuit breaker. The power supply green light should indicate it is on. Measure DC voltage of charger at the battery quick connects terminals. DC voltage must be between 13.3 and 13.55 volts.

Normal operation of DC circuit is from the DC charger with battery fully charged in float stage.

If battery is suspect or questionable, replace immediately as this is an insignificant maintenance cost. The manufacturer of the battery is Powersonic Battery and the manufacturer for the power supply is PULS. Contact GEXPRO (Formally GE Supply) 414-527-6600 or 800-279-7925 for battery or charger replacement.

Battery catalog number: PS1270
Charger catalog number ML-15.121

Specific battery maintenance and recharge/discharge information can be found in attached Powersonic documents and on the Powersonic web site (below).

Battery life is 2 years at which time it must be replaced. Initial date code on top of battery is read as (mm/dd/y/c). Where c = Q.C. inspector code.

Powersonic Battery can be located at: www.power-sonic.com.

Battery parts or assistance contact BEVCO Engineering, 262-820-2400
or
GE Healthcare battery FRU 2276547-14

Refer to sections 4-3 and 4-4 for battery manufacturer's data and specifications.

4-3 Battery Data

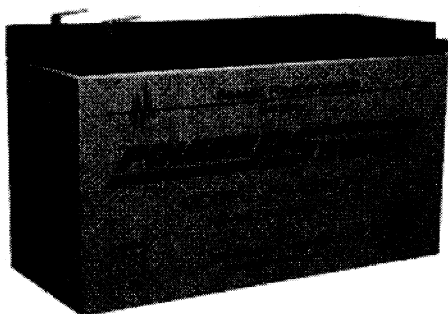


POWER SONIC

Rechargeable Sealed Lead-Acid Battery

PS-1270

12 Volt 7.0 Amp. Hrs.



Features:

- Absorbent Glass Mat (AGM) technology for superior performance.
- Valve regulated, spill proof construction allows safe operation in any position.
- Power/volume ratio yielding unrivaled energy density.
- Rugged plastic case and cover
- Approved for transport by air. D.O.T., I.A.T.A., F.A.A. and C.A.B. certified.
- U.L. recognized under file number MH 20845.

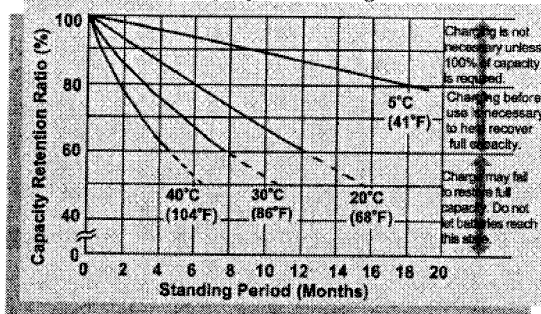


PERFORMANCE SPECIFICATIONS

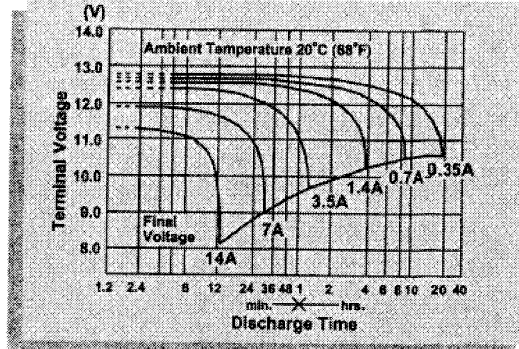
Nominal Voltage.....	12 volts (6 cells in series)
Nominal Capacity.....	
20 hour rate (350mA to 10.50 volts).....	7.0 A.H.
10 hour rate (650mA to 10.50 volts).....	6.5 A.H.
5 hour rate (1.2A to 10.20 volts).....	6.0 A.H.
1 hour rate (4.5A to 9.00 volts).....	4.5 A.H.
15 min rate (14A to 9.00 volts).....	3.5 A.H.
Approximate Weight.....	5.70 pounds (2.59 kg)
Energy Density (20 hour rate).....	1.43 Watt-hours/cubic inch (87.3 Watt-hours/l)
Specific Energy (20 hour rate).....	14.7 Watt-hours/pound (32.4 Watt-hours/kg)
Internal Resistance (Fully Charged Battery).....	22 milliohms (approximately)
Maximum Discharge Current (≤ 7 Min.).....	21 amperes
Maximum Short-Duration Discharge Current (≤ 10 Sec.).....	70 amperes
Terminal configurations.....	Quick disconnect tabs, "F1": 0.187" x 0.032", mate with AMP. INC. FASTON "187" Quick disconnect tabs, "F2": 0.250" x 0.032", mate with AMP. INC. FASTON "250"
Shelf Life — % of nominal capacity at 68° F (20° C)	
1 Month.....	97%
3 Months.....	91%
6 Months.....	83%
Operating Temperature Range	
Charge.....	-4°F (-20°C) to 122°F (50°C)
Discharge.....	-4°F (-20°C) to 140°F (60°C)
Case.....	ABS Plastic

BATTERY MANUFACTURER DATA SHEET
ILLUSTRATION 4-1

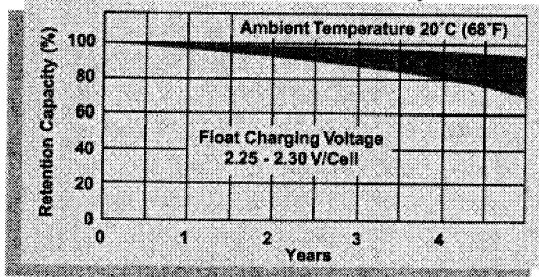
Shelf Life and Storage



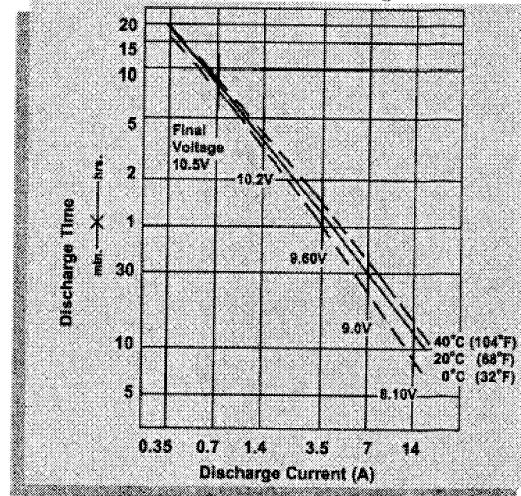
Discharge Characteristics



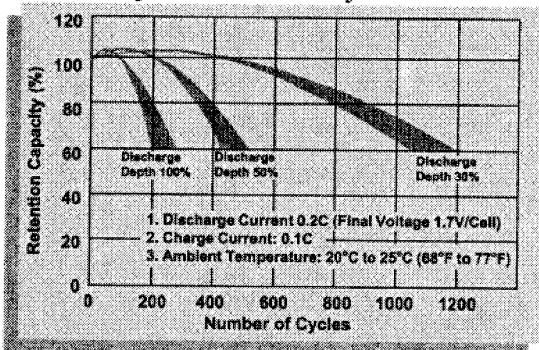
Life Characteristics in Stand-By Use



Discharge Time vs. Discharge Current



Life Characteristics in Cyclic Use



CHARGING

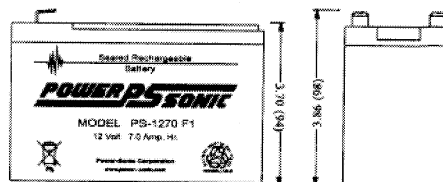
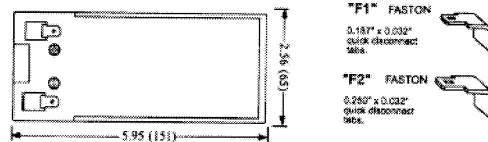
Cycle Applications: Limit initial current to 2.1A. Charge until battery voltage (under charge) reaches 14.40 to 14.70 volts at 68°F (20°C). Hold at 14.40 to 14.70 volts until current drops to approximately 70mA. Battery is fully charged under these conditions, and charger should either be disconnected or switched to "float" voltage.

"Float" or "Stand-By" Service: Hold battery across constant voltage source of 13.50 to 13.80 volts continuously. When held at this voltage, the battery will seek its own current level and maintain itself in a fully charged condition.

NOTE: Due to the self-discharge characteristics of this type of battery, it is imperative that they be charged after 6-9 months of storage, otherwise permanent loss of capacity might occur as a result of sulfation.

Physical Dimensions: in. (mm)

Terminals



Tolerances are ± 0.04 in. (± 1 mm) and ± 0.08 in. (± 2 mm) for height dimensions. All data subject to change without notice.

POWER SONIC

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FM39170

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customer-service@power-sonic.com

BATTERY MANUFACTURER DATA SHEET
ILLUSTRATION 4-2

4-4 Battery Manufacturer's Material Safety Data Sheet



MATERIAL SAFETY DATA SHEET

**PS, PSH, PSG, PG and Power Sport Series
Valve Regulated (VRLA) Batteries
Absorbed Electrolyte (AGM)**

Section 1 - Product Identification

Manufacturers Name	Emergency Telephone Numbers:	
Power-Sonic Corporation, 7550 Panasonic Way San Diego, CA 92154	CHEMTREC (Domestic):	(800) 424-9300
	CHEMTREC (International):	(703) 527-3887
	Telephone Number for Information	
	Power-Sonic Corporation:	(619) 661-2020
	Date Issued:	January 2, 2007
<i>The information contained within is provided as a service to our customers and is for their information only. The information and recommendations set forth herein are made in good faith and are believed to be accurate at the date compiled. Power-Sonic Corporation makes no warranty expressed or implied.</i>		

Section 2 - Hazardous Ingredients/Identity Information

Components	CAS Number	Approx Wt. %	OSHA PEL (µg/m³)	ACGIH TLV (µ/m³)	NIOSH (µ/m³)
Inorganic Lead/Lead Compounds	7439-92-1	65%-75%	50	150	10
Tin	7440-31-5	<0.5%	2000	2000	N/A
Calcium	7440-70-2	<0.1%	N/A	N/A	N/A
Electrolyte: Dilute sulfuric Acid	7664-93-9	14-20%	1000	1000	1000
Fiberglass Separator	-	5%	N/A	N/A	N/A
Case Material: Acrylonitrile Butadiene Styrene (ABS)	9003-56-9	5-10%	N/A	N/A	N/A

Inorganic lead and electrolyte (sulfuric acid) are the main components of every Valve Regulated Lead Acid battery supplied by Power-Sonic Corporation. Other ingredients may be present dependent upon the specific battery type. For additional information contact Power-Sonic Corporation Technical Department.

Section 3 - Physical/Chemical Characteristics

Components	Density	Melting Points	Solubility (H2O)	Odor	Appearance
Lead	11.34	621 F°	None	None	Silver-Gray
Lead Sulfate	6.20	1950 F°	40mg/l(60 F°)	None	White Powder
Lead Dioxide	9.40	554 F°	None	None	Brown Powder
Sulfuric Acid	About 1.30	203-240 F°	100%	Sharp penetrating pungent	Clear Colorless Liquid
Fiberglass Separator	N/A	N/A	Slight	None	White Fibrous
Case Material: Acrylonitrile Butadiene Styrene (ABS)	N/A	N/A	None	None	Solid

Section 4 – Flammability Data

Components	Flashpoint	Explosive Limit	Comments
Lead and Sulfuric Acid	None	None	None
Hydrogen		LEL = 4.1%	Sealed batteries can emit hydrogen if overcharged (float voltage > 2.40 VPC)
Fiberglass Separator	N/A	N/A	Toxic vapors may be released. In case of fire, wear self contained breathing apparatus
Acrylonitrile Butadiene Styrene (ABS)	None	N/A	Temp over 527°F (300°C) may release combustible gases. In case of fire, wear self contained breathing apparatus

Section 5 - Reactivity Data

Stability	Unstable		Conditions to Avoid
	Stable	X	Prolonged overcharge on high current, ignition sources. Sulfuric acid remains stable at all temperatures
Incompatibility (Materials to Avoid)			
Sulfuric acid: Contact with combustibles and organic materials may cause fire and explosion. Also reacts violently with strong reducing agents, metals, sulfur trioxide gas, strong oxidizers, and water. Contact with metals may produce toxic sulfur dioxide fumes and may release flammable hydrogen gas. Lead Compounds: Avoid contact with strong acids, bases, halides, halogenates, potassium nitrate, permanganate, peroxides, nascent hydrogen, and reducing agents.			
Hazardous Decomposition or Byproducts			
Sulfuric acid: Sulfur trioxide, carbon monoxide, sulfuric acid mist, sulfur dioxide, and hydrogen sulfide. Lead Compounds: High temperatures above the melting point are likely to produce toxic metal fume, vapor, or dust; contact with strong acid or base or presence of nascent hydrogen may generate highly toxic arsine gas. Hazardous Polymerization.			
Polymerization: Sulfuric acid will not polymerize			
Decomposition Products: Sulfuric Dioxide, Trioxide, Hydrogen Sulfide, Hydrogen.			
Conditions to Avoid: Prohibit smoking, sparks, etc. from battery charging area. Avoid mixing acid with other chemicals.			

Section 6 - Health Hazard Data

Routes of Entry
Sulfuric acid: Harmful by all routes of entry Lead compounds: Hazardous Exposure can occur only when product is heated, oxidized, or otherwise processed or damaged to create dust, vapor or fume.
Inhalation
Sulfuric Acid: Breathing sulfuric acid vapors and mists may cause severe respiratory problems. Lead Compounds: Dust or fumes may cause irritation of upper respiratory tract or lungs. Fiberglass Separator: Fiberglass is an irritant to the upper respiratory tract, skin and eyes. For exposure up to 10F[°]/ use MSA Comfoil with type H filter. Above 10F use Ultra Twin with type H filter. This product is not considered carcinogenic by NTP or OSHA.
Skin Contact
Sulfuric acid: Severe irritation, burns, cornea damage, and possible blindness. Lead Compounds: May cause eye irritation

Continued on next page

Ingestion Sulfuric acid: May cause severe irritation of the mouth, throat, esophagus, and stomach. Lead Compounds: May cause abdominal pain, nausea, vomiting, diarrhea, and severe cramping. Acute ingestion should be treated by a physician.
Eye Contact Sulfuric acid: Severe irritation, burns, cornea damage and possible blindness. Lead Compounds: May cause eye irritation.
Acute Health Hazards Sulfuric acid: Severe skin irritation, burns, damage to cornea may cause blindness, upper respiratory irritation. Lead Compounds: May cause abdominal pain, nausea, headaches, vomiting, loss of appetite, severe cramping, muscular aches and weakness, and difficulty sleeping. The toxic effects of lead are cumulative and slow to appear. It affects the kidneys, reproductive and central nervous systems. The symptoms of lead overexposure are listed above. Exposure to lead from a battery most often occurs during lead reclamation operations through the breathing or ingestion of lead dust or fumes.
Chronic Health Hazards Sulfuric acid: Possible scarring of the cornea, inflammation of the nose, throat and bronchial tubes, possible erosion of tooth enamel. Lead Compounds: May cause anemia, damage to kidneys and nervous system, and damage to reproductive system in both males and females.
Carcinogenicity Sulfuric acid: The National Toxicological Program (NTP) and The International Agency for Research on Cancer (IARC) have classified strong inorganic acid mist containing sulfuric acid as a Category 1 carcinogen, a substance that is carcinogenic to humans. The ACGIH has classified strong inorganic acid mist containing sulfuric acid as an A2 carcinogen (suspected human carcinogen). These classifications do not apply to liquid forms of sulfuric acid or sulfuric acid solutions contained within a battery. Inorganic acid mist (sulfuric acid mist) is not generated under normal use of this product. Misuse of the product, such as overcharging, may result in the generation of sulfuric acid mist. Lead Compounds: Human studies are inconclusive regarding lead exposure and an increased cancer risk. The EPA and the International Agency for Research on Cancer (IARC) have categorized lead and inorganic lead compounds as a B2 classification (probable/possible human carcinogen) based on sufficient animal evidence and inadequate human evidence.
Medical Conditions Generally Aggravated by Exposure Inorganic lead and its compounds can aggravate chronic forms of kidney, liver, and neurological diseases. Contact of battery electrolyte (acid) with the skin may aggravate skin diseases such as eczema and contact dermatitis. Overexposure to sulfuric acid mist may cause lung damage and aggravate pulmonary conditions.
Emergency and First Aid Procedures <u>Inhalation</u> Sulfuric acid: Remove to fresh air immediately. If breathing is difficult, give oxygen Lead Compounds: Remove from exposure, gargle, wash nose and lips, consult physician <u>Ingestion</u> Sulfuric acid: Do not induce vomiting, consult a physician immediately. Lead Compounds: Consult a physician immediately <u>Eyes</u> Sulfuric acid: Flush immediately with water for 15 minutes, consult a physician. Lead Compounds: Flush immediately with water for 15 minutes, consult a physician <u>Skin</u> Sulfuric acid: Flush with large amounts of water for at least 15 minutes, remove any contaminated clothing. If irritation develops seek medical attention. Lead Compounds: Wash with soap and water.

Continued on next page

Section 7 - Precautions for Safe Handling and Use

Steps to be Taken in Case Material is Released or Spilled There is no release of material unless the case is damaged or battery is misused/overcharged. If release occurs stop flow of material, contain/absorb all spills with dry sand, earth, or vermiculite. Do not use combustible materials. Neutralize spilled material with soda ash, sodium bicarbonate, lime, etc. Wear acid-resistant clothing, boots, gloves, and face shield. Dispose of as hazardous waste. Do not discharge acid to sewer.
Waste Disposal Method Spent Batteries – send to secondary lead smelter for recycling. Follow applicable federal, state and local regulations. Neutralize as in preceding step. Collect neutralized material in sealed container and handle as hazardous waste as applicable. A copy of this MSDS must be supplied to any scrap dealer or secondary lead smelter with the battery.
Precautions to be Taken in Handling and Storing Store batteries in a cool, dry, well ventilated area that are separated from incompatible materials and any activities which may generate flames, sparks, or heat. Keep all metallic articles that could contact the negative and positive terminals on a battery and create a short circuit condition.
Electrical Safety Due to the battery's low internal resistance and high power density, high levels of short circuit current can be developed across the battery terminals. Do not rest tools or cables on the battery. Use insulated tools only. Follow all installation instructions and diagrams when installing or maintaining battery systems.
Fiberglass Separator Fiberglass is an irritant to the upper respiratory tract, skin and eyes. For exposure up to 10F°/ use MSA Comfoll with type H filter. Above 10F use Ultra Twin with type H filter. This product is not considered carcinogenic by NTP or OSHA.

Section 8 - Control Measures

Respiratory Protection None required under normal conditions. If battery is overcharged and concentrations of sulfuric acid are known to exceed PEL use NIOSH or MSH approved respiratory protection.	
Engineering Controls Store and handle batteries in a well ventilated area. If mechanical ventilation is used, components must be acid resistant.	
Protective Gloves None needed under normal conditions. If battery case is damaged use rubber or plastic elbow length gauntlets.	Eye Protection None needed under normal conditions. If handling damaged or broken batteries use chemical splash goggles or face shield.
Other Protective Clothing or Equipment None needed under normal conditions. In case of damaged or broken battery use an acid resistant apron. Under severe exposure or emergency conditions wear acid resistant clothing.	
Work Hygienic Practices Handle batteries carefully to avoid damaging the case. Do not allow metallic articles to contact the battery terminals during handling. Avoid contact with the internal components of the battery.	

Continued on next page

4-5 Battery and Power Supply Specifications

Refer to Tables 4-1 and 4-2 for manufacturer's general specifications for the POWERSONIC PS1270 battery and the Puls Power Supply.

TABLE 4-1
GENERAL BATTERY SPECIFICATIONS

Power Sonic Model #	Nominal Voltage (V)	Nominal Capacity 20 hr rate (mA)	Discharge Current 20 hr rate (mA)	Dimensions								Approx. Weight		Standard Terminal Type
				Length		Width		Height		H.O.T.				
				in.	mm.	in.	mm.	in.	mm.	in.	mm.	lbs.	kg.	
PS1270	12	7.0	350	5.95	151	2.56	65	3.7	94	3.86	98	5.7	2.6	T2

4-6 POWER SUPPLY SPECIFICATIONS

TABLE 4-2

PULS Model	Nom. Voltage (V)	Voltage Range (VDC)	Nominal Current (A)	Type of Charger
ML-15.121	12	12-15	1.3	Regulated Input rated 100-240VAC



A Output Terminals

Screw terminals,
Dual terminals for the negative pole allows an easy earthing of the output voltage
+ Positive output
- Negative (return) output

B Input Terminals

Screw terminals
L Phase (Line) input
N Neutral conductor input
⊕ PE (Protective Earth) input

C DC-on LED (green)

On when the voltage on the output terminals is > 10.5V

D Output voltage potentiometer

Turn to set the output voltage. Factory set: 12.0V

Battery Recycling

Below is the web address for Environmental Safety and Health on-line. Enter this web address. Scroll down to "Where to Recycle Your Batteries". Here you will find two links to locate local recycling centers in California or all other states and Provinces in the US and Canada.

<http://www.ehso.com/ehshome/batteries.php>

5 PARTS LIST

5-1 Renewable Parts

Renewable parts for the DISCOVERY MR750 or W450 Main Disconnect Panels.

Table 5-1
RENEWABLE PARTS AND GEHC FIELD REPLACABLE PART NUMBERS (FRU)

			Manufacturer Part Number		
Part	Volts	Amps	Bussmann	Mersen- Ferraz	GEHC-FRU
Fuse FU1	380,480	1.5	FNQ-R-1-1/2	ATQR1-1/2	5161287-2
Fuse FU2	480	60	LPJ60SP	AJT-60	5161287-3
Fuse FU3	12VDC	2	LP-CC-2	ATDR-2	
Red PB Operator*			P9XPNRG		
Contact Block*			P9B01VN		
Name Plate*			Emergency Off		
SS Wall Plate*			BEVCO SS 22mm, 1 hole		
Green PB Operator			P9SPNVG		5161287-4
Pilot Lamp (Green)			P9PDNVJLV		
Battery	12 DC	7.0Ah	PS1270		2276547-14
Battery Charger	12 DC	1.3A	ML-15.121		
* = Component parts of additional remote emergency OFF pushbutton					

All of the above parts are available from GEXPRO (Formally GE Supply) Milwaukee (414)-527-6600 or 1-800-279-7925 Items marked as FRUS are also available from GE Healthcare.

5-2 Replacement Parts

See Table 5-2 for replacement parts for the DISCOVERY MR750 and W450 System Main Disconnect Panel.

Table 5-2
LIST FOR REPLACEMENT PARTS

ITEM	Catalog number	Description	Quantity
Battery	PS1270	BATTERY, 12VDC, 7Ah, .250 Tabs	1.00
Battery	SD-385	BATTERY BRACKET (VERTICAL MTG)	1.00
C1	CL06A311MJ	CONTACTOR, 48A, IEC, NON-REV	1.00
CB 2, CB3	TCAL18	LUG KIT, #12-3/0	9.00
CB1	SC4L	CABLE, 48", OPERATOR FOR DISC.	1.00
CB1	SCH1	OPERATING HANDLE	1.00
CB1	SCOM1EF	OPERATOR MECHANISM	1.00
CB1	SFHA36AT0250	FRAME,CIRCUIT BREAKER,250A	1.00
CB1	SFMSK1	MOUNTING KIT,CIRCUIT BREAKER	1.00
CB1	SRPF250A200	PLUG,RATING,200A	1.00
CB1	TCAL28PD3	Distribution lug for SFHA	1.00
CB1	TCAL29	LINE LUG, #8-350MCM	3.00
CB1, CB2	SEMSK3	MOUNTING KIT	2.00
CB1,CB2	SEPLD	PADLOCK ATTACHMENT	2.00
CB2	SAUV1	UV TRIP	1.00
CB2	SEHA36AT0150	CIRCUIT BREAKER, 150 AMP FRAME	1.00
CB2	SRPE150A150	RATING PLUG, 150AMP	1.00
CB2	54106	LUG, COMPRESSION,4GA,1/4" HOLE	6.00
CB3	SEHA36AT0060	CIRCUIT BREAKER FRAME,60A	1.00
CB3	SRPE60A50	RATING PLUG,50A	1.00
CB4	EP101ULC03	CIRCUIT BREAKER, 1P, 3A	1.00
CB5	EP101ULC01	CIRCUIT BREAKER,1-POLE,1-AMP	1.00
CR1	RY4S-UDC12V	RELAY,12VDC,4PDT,	1.00
CR1	SY4S-05	SOCKET, RELAY, DIN MOUNT	1.00
CR1	SY4S-51F1	SPRING, HOLD DOWN FOR RELAY	1.00
CR2	451WFPR1D01	RELAY, PLUGGABLE, 12VDC COIL	1.00
FU1	CHCC2DU	FUSE BLOCK,2P,600V,30A,REJECT	1.00
FU1	FNQR1.5	FUSE	2.00
FU2	AJT60	FUSE, 600V, 60A, CLASS J	3.00

FU2	JT60060	FUSE BLOCK, 1P, 60A, CLASS J	3.00
FU3	CCMR2	FUSE,2A,TIME DELAY,600V,REJECT	1.00
FU3	CHCC1DU	FUSE BLOCK, 1P, 30A, CLASS CC	1.00
G/PE	ADR25	GROUND LUG, #6-250MCM, 455A	2.00
G/PE	PK12GTA	GROUND BUSS, COPPER	1.00
G/PE SHIELD	101010	TERMINAL, GROUND	2.00
LT1	P9PDNVJLV	LED LAMP, 120vAC, GREEN	1.00
LT1	P9SLVD	PILOT LIGHT, GREEN	1.00
NEUTRAL	PDBFS303	DISTRIBUTION BLOCK,	1.00
PB1	P9ASBGV028	CAP, GRN, FLUSH, IEC START SYMBOL	1.00
PB1	P9B10VN	CONTACT BLOCK, NO, IEC	1.00
PB1	P9SPNVG	SWITCH, PUSHBUTTON, GREEN, FLUSH	2.00
PB1,2 LT1	P9ASTBS	NAMEPLATE	3.00
PB2	P9ASBGR029	CAP, PUSHBUTTON, INTL STOP	1.00
PB2	P9B01VN	CONTACT BLOCK, NC, IEC	1.00
PB2	P9XPNRS	PB 22MM RED, ROHS COMPLIANT	2.00
PB3,PB4	BEVCO 22.5MM wall plate	WALL PLATE, SS, P9, 22.5mm HOLE	2.00
PB3,PB4	P9SPNRG-P9B01VN-EPO- NP	REMOTE PUSHBUTTON, EMERGENCY OFF, NC	2.00
PNPT	INNOVA LABEL	LABEL, SYSTEM MAIN DISC PANEL	1.00
PS1	ML15.121	POWER SUPPLY, 15V, 1.3A, 100-240V	1.00
T1	9T58E0506	TRANSFORMER, 208/400:120, 250VA	1.00
TERMINALS	102010	TERMINAL BLOCK, WDU 4, BEIGE	10.00
TERMINALS	105000	END BARRIER, WAP 2.5-10, BEIGE	3.00
TERMINALS	106120	END STOP, WEW 35/2	8.00

Revision history

REV	DATE	AUTHOR	PRIMARY REASONS FOR CHANGE
1	9/15/2008	L Hernandez	Editorial revisions, added FRU numbers
2	11/22/2008	L Hernandez	Removed label, added terminal strip picture
3	1/29/2009	L Hernandez	Changed picture 1-1-3-2
4	8/26/2014	L Hernandez	DRAFT-Updated photos, drawings, deleted 200A fuses, now RoHS compliant chg remote EPO to P9, BEVCO for parts.
5	3/17/2015	L Hernandez J Benedick	Updated drawing, pictures to Rev 9 of PSP. Removed Draft. Added location information.
6	3/24/2015	L Hernandez	Changed weight, remote EPO PB cat number
7	1/19/2016	L Hernandez	Added annual tightening of bolts by electrician
7.1	5-9-2018	L Hernandez	Added EPO connection instructions, added CB1 adjustment reference, better cover photo.
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